

A S M Sharif Mahmud Sazid

Software Engineer • AWS Certified Solutions Architect – Associate

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🌐 [sharif-mahmud-sazid](#) | 🏠 [GitHub](#) | 🌐 [Portfolio](#)

Professional Summary

Software Engineer and AWS Certified Solutions Architect – Associate with hands-on experience building full-stack applications, designing REST APIs, implementing CI/CD pipelines, and deploying scalable systems using React, Next.js, Node.js, PostgreSQL, Docker, Linux, and AWS. Comfortable rapidly learning new technologies and using AI-assisted development tools while rigorously validating correctness, security, and code quality.

Education

BSc in Computer Science and Engineering

2022–2026

Shahjalal University of Science and Technology (SUST), Sylhet

- Relevant coursework: Data Structures & Algorithms, Software Engineering, DBMS, Operating Systems, Networking & Security, Cloud Computing, Machine Learning

Technical Skills

Languages & Frameworks: TypeScript, JavaScript, Python, Java, C++, OOP, SOLID, React, Next.js, Node.js, Express.js

Cloud & AWS: EC2, S3, Lambda, IAM, VPC, RDS, ALB, ECS, ECR, Route 53, CloudWatch, CloudTrail, API Gateway, DynamoDB

DevOps & Infrastructure: Git, Docker, Docker Compose, GitHub Actions, CI/CD, Linux, Bash, Nginx

AI Development: GitHub Copilot, Cursor, Claude, Antigravity, AI-assisted development, prompt engineering

Databases & ORMs: PostgreSQL, MySQL, MongoDB, Firebase, Prisma ORM, Drizzle ORM, pgvector

AI/ML Research: TensorFlow, Deep Learning, CNNs, GNNs, RAG, Embeddings, Vector Search, LLM Applications

Selected Projects

ResearchRAG — AI Research Paper Assistant [🔗](#)

2026

Next.js 16, TypeScript, Python, PostgreSQL, pgvector, Cohere, Supabase, Docker

- Designed and implemented a Retrieval-Augmented Generation (RAG) system for querying academic PDFs with cited answers using HyDE retrieval, Cohere embeddings, pgvector search, and Cohere Rerank v3
- Fault-tolerant four-tier LLM waterfall (Cohere → Gemini → DeepSeek/Llama → GPT-4o Mini) with automatic failover

SUST Research Hub — Academic Research Platform [🔗](#)

2025

Next.js 16, TypeScript, PostgreSQL, Prisma ORM, Cloudinary CDN, Docker

- Full-stack academic platform for managing theses, datasets, and publications, with 40+ relational tables and automated workflow notifications
- Multi-role authentication for students, supervisors, and admins; Dockerized with Cloudinary CDN for file storage at scale

HomeSync — Smart Home IoT Dashboard [🔗](#)

2025

React 19, TypeScript, Node.js, Express, PostgreSQL, Socket.io, Docker, Nginx, AWS EC2

- Designed RESTful APIs and a full-stack “Digital Twin” IoT dashboard with real-time Socket.io updates and a simulation engine for live sensor data
- Deployed to AWS EC2 (ap-south-1) with Docker Compose, Nginx, and a GitHub Actions CI/CD pipeline for auto-deploy on push

ConsultBook — Expert Consultation SaaS Platform [🔗](#)

2024

Next.js 15, TypeScript, PostgreSQL, Drizzle ORM, Stripe, Server-Sent Events, Docker

- Full-stack SaaS marketplace for 1:1 expert consultations with smart scheduling, timezone support, and Stripe payments/payouts
- Real-time messaging with SSE notifications and dual-role dashboards with earnings analytics; Dockerized for deployment

Research: Hierarchical Deep Learning for Microorganism Classification [🔗](#)

2026

TensorFlow, ResNet50, ResNet152V2, Transfer Learning

- Designed a two-stage hierarchical CNN (ResNet50 → ResNet152V2 via confidence-based routing) for 38-class microorganism classification, reaching 98.40% accuracy and 100% F1 on bacterial detection
- Curated and cleaned a 7,000+ image dataset from three public sources; validated the approach against single-stage baselines with McNemar’s significance testing and ablation studies

Technical Achievements

- **Competitive Programmer** — Codeforces *Specialist* (Max Rating: 1567)
- **LeetCode** — *Knight* (Contest Rating: 1864)
- **Open Source** — Active GitHub contributor, 25+ public repositories demonstrating self-directed learning across multiple technologies

Certifications

- AWS Certified Solutions Architect – Associate (June 2026) [🔗](#)
- AWS Academy — Cloud Developing [🔗](#)
- AWS Academy — Cloud Foundations [🔗](#)